

DERIVED CATEGORIES AND DERIVED FUNCTORS

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Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor between abelian categories. In this note, we define the right derived functor $RF : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$, and study its fundamental properties. To simplify matters, we will assume that $\mathcal{A}$ has enough injective objects.

We assume that the reader is familiar with the notions of additive categories and abelian categories. While working within a given abelian category, we will occasionally make arguments (e.g. diagram chases) where we take for granted the fact that our objects have elements. Although in most cases such arguments can be extended to general abelian categories without much trouble, we will not take the time to explain how this is done.

1. THE HOMOTOPY CATEGORY

Let $\mathcal{A}$ be an additive category, and let $\text{Com}(\mathcal{A})$ denote the category of complexes of $\mathcal{A}$.

- Definition of cochain homotopy (or definition of homotopic to 0).
- Homotopy is an equivalence relation. Definition of $K(\mathcal{A})$.
- Quasi-isomorphism makes sense in $K(\mathcal{A})$.

2. MAPPING CONES AND TRIANGLES

Let $\mathcal{A}$ be an additive category. In this section, we discuss the triangulated structure on the homotopy category $K(\mathcal{A})$. Since we won't make any essential use of the theory of triangulated categories in our treatment of derived functors, we will treat it as a convenient conceptual language, without worrying too much about the more formal aspects. In particular, we will omit the full definition of triangulated category, and the verification that $K(\mathcal{A})$ is triangulated.

To define a triangulated structure on an additive category, we have to specify two things:

- i) an additive autoequivalence, called a *shift functor*;
- ii) a class of *distinguished triangles*;

which together must satisfy a short list of axioms. In the case of $K(\mathcal{A})$, the shift functor is denoted by $[1]$; its n th iterate is denoted by $[n]$. It is defined on an object $A^\bullet \in K(\mathcal{A})$ by

$$A^\bullet[1]^i := A^{i+1}, \quad d_{A[1]} := -d_A,$$

and on a morphism $f : A^\bullet \rightarrow B^\bullet$ by $f[1]^i := f^{i+1}$. To define the distinguished triangles in $K(\mathcal{A})$, we have to introduce mapping cones.

Definition 2.1. Suppose $f : A^\bullet \rightarrow B^\bullet$ is a morphism of complexes. Its *mapping cone* is the complex $C(f)$ defined by

$$C(f)^i := A^{i+1} \oplus B^i, \quad d_{C(f)}^i := \begin{bmatrix} -d_{A^\bullet}^{i+1} & 0 \\ f^{i+1} & d_{B^\bullet}^i \end{bmatrix}$$

There are natural morphisms of complexes $\tau : B^\bullet \rightarrow C(f)$ and $\pi : C(f) \rightarrow A^\bullet[1]$, fitting in a short exact sequence

$$0 \longrightarrow B \xrightarrow{\tau} C(f) \xrightarrow{\pi} A^\bullet[1] \longrightarrow 0. \quad (2.1)$$

Note that while the other composites $\tau \circ f$ and $f[1] \circ \pi$ are not necessarily zero, they are *homotopic* to zero, as one easily checks.

A useful property of mapping cones is that they detect quasi-isomorphisms:

Lemma 2.2. *Let $f : A^\bullet \rightarrow B^\bullet$ be a map of complexes. Then f is a quasi-isomorphism if and only if its mapping cone $C(f)$ is an exact complex.*

Proof. Consider the long exact sequence in cohomology of the short exact sequence (2.1), and observe that the connecting homomorphism $H^i(A^\bullet[1]) \rightarrow H^{i+1}(B^\bullet)$ is just $H^{i+1}(f)$. $\square$

Definition 2.3. A *triangle* is a sequence in $K(\mathcal{A})$ of the form

$$A_1^\bullet \xrightarrow{u} A_2^\bullet \xrightarrow{v} A_3^\bullet \xrightarrow{w} A_1^\bullet[1],$$

where the composites $v \circ u$, $w \circ v$ and $u[1] \circ w$ are the zero morphisms. A triangle is *distinguished* if it is isomorphic to a triangle of the form

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{\tau} C(f) \xrightarrow{\pi} A^\bullet[1].$$

This of course means that there is a commutative diagram in $K(\mathcal{A})$:

$$\begin{array}{ccccccc} A_1^\bullet & \xrightarrow{u} & A_2^\bullet & \xrightarrow{v} & A_3^\bullet & \xrightarrow{w} & A_1^\bullet[1] \\ \downarrow \varphi & & \downarrow \psi & & \downarrow \chi & & \downarrow \varphi[1] \\ A^\bullet & \xrightarrow{f} & B^\bullet & \xrightarrow{\tau} & C(f) & \xrightarrow{\pi} & A^\bullet[1], \end{array}$$

where the vertical maps are isomorphisms.

The distinguished triangles in $K(\mathcal{A})$ satisfy several natural properties, which when abstracted, comprise the axioms of a triangulated category. For example, any triangle of the form

$$A^\bullet \xrightarrow{\text{id}} A^\bullet \longrightarrow 0 \longrightarrow A^\bullet[1]$$

is distinguished, and any commutative diagram of the form

$$\begin{array}{ccccccc} A_1^\bullet & \xrightarrow{f_1} & B_1^\bullet & \longrightarrow & C(f_1) & \longrightarrow & A_1^\bullet[1] \\ \downarrow & & \downarrow & & \vdots & & \downarrow \\ A_2^\bullet & \xrightarrow{f_2} & B_2^\bullet & \longrightarrow & C(f_2) & \longrightarrow & A_2^\bullet[1], \end{array}$$

can be completed with a dashed arrow. Another important property is the fact that the class of distinguished triangles is closed under ‘rotations’: if a triangle

$$A_1^\bullet \xrightarrow{u} A_2^\bullet \xrightarrow{v} A_3^\bullet \xrightarrow{w} A_1^\bullet[1]$$

is distinguished, then so too are the rotated triangles

$$A_2^\bullet \xrightarrow{-v} A_3^\bullet \xrightarrow{-w} A_1^\bullet[1] \xrightarrow{-u[1]} A_2^\bullet[1]$$

and

$$A_3^\bullet[-1] \xrightarrow{-w[1]} A_1^\bullet \xrightarrow{-u} A_2^\bullet \xrightarrow{-v} A_3^\bullet.$$

This is a consequence of the following lemma:

Lemma 2.4. *Let $f : A^\bullet \rightarrow B^\bullet$ be a morphism of complexes, and consider the mapping cone $C(\tau)$ of the inclusion $\tau : B^\bullet \rightarrow C(f)$. There exists a morphism of complexes $g : A^\bullet[1] \rightarrow C(\tau)$, which is an isomorphism in $K(\mathcal{A})$, and which makes the following diagram commutative in $K(\mathcal{A})$:*

$$\begin{array}{ccccccc} B^\bullet & \xrightarrow{\tau} & C(f) & \xrightarrow{\pi} & A^\bullet[1] & \xrightarrow{-f} & B^\bullet[1] \\ \parallel & & \parallel & & \downarrow g & & \parallel \\ B^\bullet & \xrightarrow{\tau} & C(f) & \xrightarrow{\tau\tau} & C(\tau) & \xrightarrow{\pi\tau} & B^\bullet[1]. \end{array}$$

Proof. First, note that the mapping cone $C(\tau)$ is given in degree i by

$$C(\tau)^i = B^{i+1} \oplus C(f)^i = B^{i+1} \oplus A^{i+1} \oplus B^i,$$

with differential

$$d_{C(\tau)}^i = \begin{bmatrix} -d_B^{i+1} & 0 & 0 \\ 0 & -d_A^{i+1} & 0 \\ \text{id}_{B^{i+1}} & f^{i+1} & d_B^i \end{bmatrix}.$$

Define g by the matrix

$$g^i := \begin{bmatrix} -f^{i+1} \\ \text{id}_{A^{i+1}} \\ 0 \end{bmatrix}.$$

It's then easy to check that g is a morphism of complexes. Define $p : C(\tau) \rightarrow A^\bullet[1]$ by projection onto the middle factor. Then p is a morphism of complexes such that $pg = \text{id}_{A^{i+1}}$, while gp is given by the matrix product

$$\begin{bmatrix} -f^{i+1} \\ \text{id}_{A^{i+1}} \\ 0 \end{bmatrix} \begin{bmatrix} 0 & \text{id}_{A^{i+1}} & 0 \end{bmatrix} = \begin{bmatrix} 0 & -f^{i+1} & 0 \\ 0 & \text{id}_{A^{i+1}} & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

We claim that gp is homotopic to the identity on $C(\tau)$; that is, $\text{id}_{C(\tau)} - gp$ is homotopic to zero. If we define $h^i : C(\tau)^i \rightarrow C(\tau)^{i-1}$ by

$$h^i := \begin{bmatrix} 0 & 0 & \text{id}_{B^i} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

we find that

$$d_{C(\tau)}^{i-1} h^i + h^{i+1} d_{C(\tau)}^i = \begin{bmatrix} \text{id}_{B^{i+1}} & f^{i+1} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \text{id}_{B^i} \end{bmatrix}$$

— this is $\text{id}_{C(\tau)}^i - (gp)^i$. Therefore, $p = g^{-1}$ in $K(\mathcal{A})$.

The commutativity of the diagram is clear, apart from the fact that the middle square

$$\begin{array}{ccc} C(f) & \xrightarrow{\pi} & A^\bullet[1] \\ \parallel & & \downarrow g \\ C(f) & \xrightarrow{\tau\tau} & C(\tau). \end{array}$$

does not commute in $\text{Com}(\mathcal{A})$. However, note that $p\tau_\tau = \pi$, and since $p = g^{-1}$ in $\text{K}(\mathcal{A})$, this square does commute in $\text{K}(\mathcal{A})$. $\square$

Proposition 2.5. *The rotation of a distinguished triangle in $\text{K}(\mathcal{A})$ is distinguished.*

Proof. Given a distinguished triangle,

$$A_1^\bullet \xrightarrow{u} A_2^\bullet \xrightarrow{v} A_3^\bullet \xrightarrow{w} A_1^\bullet[1]$$

the triangle

$$A_2^\bullet \xrightarrow{-v} A_3^\bullet \xrightarrow{-w} A_1^\bullet[1] \xrightarrow{-u[1]} A_2^\bullet[1]$$

is distinguished by Lemma 2.4. To see that

$$A_3^\bullet[-1] \xrightarrow{-w[1]} A_1^\bullet \xrightarrow{-u} A_2^\bullet \xrightarrow{-v} A_3^\bullet$$

is a distinguished triangle, note that

$$A_1^\bullet[-1] \xrightarrow{-u[-1]} A_2^\bullet[-1] \xrightarrow{-v[-1]} A_3^\bullet[-1] \xrightarrow{-w[-1]} A_1^\bullet$$

is distinguished, since if $f : A^\bullet \rightarrow B^\bullet$ is a morphism of complexes, $C(f)[-1]$ is the mapping cone of $-f : A[-1]^\bullet \rightarrow B[-1]^\bullet$. Then apply Lemma 2.4 twice more. $\square$

Definition 2.6. Let $\mathcal{T}$ be a triangulated category and $\mathcal{B}$ an abelian category. An additive functor $H : \mathcal{T} \rightarrow \mathcal{B}$ is called a *homological functor* if for every distinguished triangle in $\mathcal{T}$:

$$A_1 \xrightarrow{u} A_2 \xrightarrow{v} A_3 \xrightarrow{w} A_1[1],$$

the sequence

$$H(A_1) \xrightarrow{H(u)} H(A_2) \xrightarrow{H(v)} H(A_3)$$

is exact.

Because the class of distinguished triangles in a triangulated category $\mathcal{T}$ is closed under rotations, any distinguished triangle can be extended infinitely in both directions, in such a way that every consecutive triple of morphisms is a distinguished triangle:

$$\begin{array}{ccccccc} A_1[1] & \xrightarrow{-u[1]} & A_2[1] & \xrightarrow{-v[1]} & \dots & & \\ \uparrow & & & & & & \\ & \xrightarrow{w} & & & & & \\ \uparrow & & & & & & \\ A_1 & \xrightarrow{u} & A_2 & \xrightarrow{v} & A_3 & & \\ & \xrightarrow{-w[-1]} & & & & & \\ \uparrow & & & & & & \\ \dots & \xrightarrow{-u[-1]} & A_2[-1] & \xrightarrow{-v[-1]} & A_3[-1] & & \end{array}$$

Hence, any homological functor H gives rise to a long exact sequence in $\mathcal{B}$ (without the annoying signs):

$$\begin{array}{ccccccc}
 H(A_1[1]) & \xrightarrow{H(u[1])} & \dots & & & & \\
 \uparrow & & & \searrow^{H(w)} & & & \\
 H(A_1) & \xrightarrow{H(u)} & H(A_2) & \xrightarrow{H(v)} & H(A_3) & & \\
 \uparrow & & & \searrow^{H(w[-1])} & & & \\
 & & \dots & \xrightarrow{H(v[-1])} & H(A_3[-1]) & &
 \end{array} \tag{2.2}$$

Example 2.7. For any $i \in \mathbb{Z}$, the cohomology functor $H^i : K(\mathcal{A}) \rightarrow \mathbf{Ab}$ is a homological functor. To see this, note that Proposition 2.5 implies that the rotated triangle

$$A_3[-1]^\bullet \xrightarrow{-w[-1]} A_1^\bullet \xrightarrow{-u} A_2^\bullet \xrightarrow{-v} A_3^\bullet$$

is distinguished; hence $A_2^\bullet$ is isomorphic to the mapping cone of $-w[-1]$. Then considering the long exact cohomology sequence of the short exact sequence

$$0 \longrightarrow A_1^\bullet \longrightarrow C(-w[-1]) \longrightarrow A_3^\bullet \longrightarrow 0,$$

we see that

$$H^i(A_1^\bullet) \longrightarrow H^i(C(-w[-1])) \longrightarrow H^i(A_3^\bullet)$$

is exact, which gives us what we need.

Remark 2.8. Note that the long exact sequence (2.2) obtained by applying H^i to the distinguished triangle

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{\tau} C(f) \xrightarrow{\pi} A^\bullet[1]$$

is the same as the long exact cohomology sequence of (2.1). In the next section, we will see that the distinguished triangles of the derived category enjoy a much closer relationship to short exact sequences.

Example 2.9. If $\mathcal{T}$ is any triangulated category, and $B \in \mathcal{T}$ is any object, the functors $\text{Hom}(B, -) : \mathcal{T} \rightarrow \mathcal{B}$ and $\text{Hom}(-, B) : \mathcal{T}^{\text{op}} \rightarrow \mathcal{B}$ are homological functors. In the latter case, this means that if

$$A_1 \xrightarrow{u} A_2 \xrightarrow{v} A_3 \xrightarrow{w} A_1[1],$$

is a distinguished triangle in $\mathcal{T}$, then

$$\text{Hom}(A_3, B) \xrightarrow{-\circ v} \text{Hom}(A_2, B) \xrightarrow{-\circ u} \text{Hom}(A_1, B)$$

is exact. To verify this, suppose $f : A_2 \rightarrow B$ is a morphism such that $f \circ u = 0$, and consider the commutative diagram

$$\begin{array}{ccccccc}
 A_1 & \xrightarrow{u} & A_2 & \xrightarrow{v} & A_3 & \xrightarrow{w} & A_1[1] \\
 \downarrow & & \downarrow f & & & & \downarrow \\
 0 & \longrightarrow & B & \xrightarrow{\text{id}} & B & \longrightarrow & 0.
 \end{array}$$

Since the bottom row is a distinguished triangle, this diagram can be completed to a morphism of triangles with an arrow $f' : A_3 \rightarrow B$. But this means $f = f' \circ v$, as needed. The verification that $\text{Hom}(B, -)$ is a homological functor is similar.

When we introduce the derived category in the next section, the tricky part will be turn out to be defining the morphisms. The following proposition will be the key to ensuring that the definitions make sense:

Proposition 2.10. *Given a quasi-isomorphism $f : A^\bullet \rightarrow B^\bullet$ and an arbitrary morphism $g : C^\bullet \rightarrow B^\bullet$, there exists a commutative diagram in $\mathbf{K}(\mathcal{A})$:*

$$\begin{array}{ccc} C_0^\bullet & \longrightarrow & C^\bullet \\ \downarrow & & \downarrow g \\ A^\bullet & \xrightarrow{f} & B^\bullet \end{array},$$

where the map $C_0^\bullet \rightarrow C^\bullet$ is a quasi-isomorphism.

Proof. Since f is a quasi-isomorphism, Lemma 2.2 implies that the mapping cone $C(f)$ is an exact complex, and it makes sense that we would try to leverage this fact somehow. Consider the mapping cone $C(\tau \circ g)$ of $\tau \circ g$, which fits into the extended diagram

$$\begin{array}{ccccc} C(\tau \circ g)[-1] & \xrightarrow{\pi_{\tau \circ g}[-1]} & C^\bullet & \xrightarrow{\tau \circ g} & C(f) \\ & & \downarrow g & & \parallel \\ A^\bullet & \xrightarrow{f} & B^\bullet & \xrightarrow{\tau} & C(f). \end{array}$$

Since $C(f)$ is acyclic, the long exact cohomology sequence associated to the short exact sequence

$$0 \longrightarrow C(f) \longrightarrow C(\tau \circ g) \longrightarrow C^\bullet[1] \longrightarrow 0$$

shows that $\tilde{f} := \pi_{\tau \circ g}[-1]$ is a quasi-isomorphism. So setting $C_0^\bullet := C(\tau \circ g)[-1]$, we will be done if we can construct a morphism $\tilde{g} : C_0^\bullet \rightarrow A^\bullet$ which makes the diagram

$$\begin{array}{ccc} C_0^\bullet & \xrightarrow{\tilde{f}} & C^\bullet \\ \downarrow \tilde{g} & & \downarrow g \\ A^\bullet & \xrightarrow{f} & B^\bullet \end{array}$$

commute in $\mathbf{K}(\mathcal{A})$. Now there is a natural map of complexes $C_0^\bullet \rightarrow C(\tau)[-1]$, which is given in degree i by

$$\begin{bmatrix} g^i & 0 \\ 0 & \text{id}_{C(f)}^{i-1} \end{bmatrix} : C^i \oplus C(f)^{i-1} \longrightarrow B^i \oplus C(f)^{i-1},$$

and which fits into a commutative square

$$\begin{array}{ccc} C_0^\bullet & \xrightarrow{\tilde{f}} & C^\bullet \\ \downarrow & & \downarrow g \\ C(\tau)[-1] & \xrightarrow{\pi_\tau} & B^\bullet. \end{array}$$

By Lemma 2.4, the canonical projection $p : C(\tau)[-1] \rightarrow A^\bullet$ is an isomorphism, and we have a commutative triangle

$$\begin{array}{ccc} C(\tau)[-1] & \xrightarrow{\pi_\tau} & B^\bullet \\ \downarrow -p & \nearrow f & \\ A^\bullet & & \end{array}$$

So if we define $\tilde{g}$ to be the above map $C_0^\bullet \rightarrow C(\tau)[-1]$ followed by $-p$, then we have what we want. $\square$

3. THE DERIVED CATEGORY

In this section, we study the derived category $D(\mathcal{A})$ of an abelian category $\mathcal{A}$. One way to think about the derived category is in terms of the following universal property: $D(\mathcal{A})$ is equipped with a functor $\text{Com}(\mathcal{A}) \rightarrow D(\mathcal{A})$ which is universal with respect to the property of mapping quasi-isomorphisms to isomorphisms.

In view of this, we might try to construct $D(\mathcal{A})$ by somehow inverting all the quasi-isomorphisms in $\text{Com}(\mathcal{A})$. Although it is possible to make this work, it turns out to be more convenient to build $D(\mathcal{A})$ out of the homotopy category $K(\mathcal{A})$. One reason for this is that $D(\mathcal{A})$ is a triangulated category, and while we saw in the previous section that $K(\mathcal{A})$ is naturally a triangulated category, there is no reasonable triangulated structure that one can put on $\text{Com}(\mathcal{A})$.

The objects of $D(\mathcal{A})$ will be the objects of $K(\mathcal{A})$, i.e. just complexes in $\mathcal{A}$. The morphisms will be equivalence classes of certain diagrams in $K(\mathcal{A})$:

Definition 3.1. A *roof* (from $A^\bullet$ to $B^\bullet$) is a diagram in $K(\mathcal{A})$ of the form

$$\begin{array}{ccc} & C^\bullet & \\ \text{qis} \swarrow & & \searrow \\ A^\bullet & & B^\bullet, \end{array}$$

where the arrow $C^\bullet \rightarrow A^\bullet$ is a quasi-isomorphism. Two roofs

$$A^\bullet \xleftarrow{\text{qis}} C_1^\bullet \longrightarrow B^\bullet, \quad A^\bullet \xleftarrow{\text{qis}} C_2^\bullet \longrightarrow B^\bullet$$

are said to be *equivalent* if there exists a third *invertible* roof

$$C_1^\bullet \xleftarrow{\text{qis}} C^\bullet \xrightarrow{\text{qis}} C_2^\bullet$$

making the following diagram commute:

$$\begin{array}{ccccc} & & C^\bullet & & \\ & \swarrow & & \searrow & \\ & C_1^\bullet & & C_2^\bullet & \\ \swarrow & & & & \searrow \\ A^\bullet & & & & B^\bullet. \end{array}$$

Note that this relation on roofs is an equivalence relation: it is obviously reflexive and symmetric, while transitivity follows easily from Proposition 2.10.

Equivalence classes of roofs can be composed in the following way. Suppose

$$A^\bullet \xleftarrow{\text{qis}} C_1^\bullet \longrightarrow B^\bullet, \quad B^\bullet \xleftarrow{\text{qis}} C_2^\bullet \longrightarrow C^\bullet$$

are roofs. Applying Proposition 2.10 to the morphisms $C_2^\bullet \rightarrow B^\bullet$ and $C_1^\bullet \rightarrow B^\bullet$, we obtain a commutative square

$$\begin{array}{ccc} & C_0^\bullet & \\ \text{qis} \swarrow & & \searrow \\ C_1^\bullet & & C_2^\bullet \\ & \searrow \quad \swarrow & \\ & B^\bullet & \end{array}.$$

The composite is defined to be the equivalence class of the diagram

$$\begin{array}{ccccc} & & C_0^\bullet & & \\ & & \text{qis} \swarrow & \searrow & \\ & C_1^\bullet & & C_2^\bullet & \\ & \text{qis} \swarrow & & \searrow & \\ A^\bullet & & & & C^\bullet. \end{array} \tag{3.1}$$

The fact that this operation is well-defined follows from another application of Proposition 2.10.

Definition 3.2. Let $\mathcal{A}$ be an abelian category. The *derived category* $D(\mathcal{A})$ of $\mathcal{A}$ has objects

$$\text{Ob}(D(\mathcal{A})) := \text{Ob}(K(\mathcal{A})) = \text{Ob}(\text{Com}(\mathcal{A}))$$

and morphisms

$$\text{Hom}_{D(\mathcal{A})}(A^\bullet, B^\bullet) := \{\text{roofs from } A^\bullet \text{ to } B^\bullet\} / \sim,$$

where the composition is given by (3.1).

Remark 3.3. This definition hides a set-theoretic subtlety: the Hom sets in $D(\mathcal{A})$ are in general too big to be genuine sets — they are proper classes! This is disturbing, since it means that $D(\mathcal{A})$ is *not* a category, in the usual sense of the term. However, we will see later in Corollary 4.10 that the derived ‘categories’ we care about (in this note) really are categories. Therefore, we feel justified in ignoring this issue.

The derived category $D(\mathcal{A})$ has a natural additive structure. Let two morphisms $\alpha_1, \alpha_2 \in \text{Hom}_{D(\mathcal{A})}(A^\bullet, B^\bullet)$ be represented by roofs

$$A^\bullet \xleftarrow{s_1} C_1^\bullet \xrightarrow{f_1} B^\bullet, \quad A^\bullet \xleftarrow{s_2} C_2^\bullet \xrightarrow{f_2} B^\bullet.$$

To add these morphisms, we first bring them to a ‘common denominator’, as follows. By Proposition 2.10, there is a commutative diagram

$$\begin{array}{ccc} C_0^\bullet & \xrightarrow{t_1} & C_2^\bullet \\ \downarrow t_2 & & \downarrow s_2 \\ C_1^\bullet & \xrightarrow{s_1} & A^\bullet, \end{array}$$

where both t_1 and t_2 are quasi-isomorphisms. Then α_1 and α_2 are represented by

$$A^\bullet \xleftarrow{s_1 t_2} C_0^\bullet \xrightarrow{f_1 t_2} B^\bullet, \quad A^\bullet \xleftarrow{s_2 t_1} C_0^\bullet \xrightarrow{f_2 t_1} B^\bullet,$$

and we define the sum $\alpha_1 + \alpha_2$ to be the class of the roof

$$\begin{array}{ccc} & C_0^\bullet & \\ s_1 t_2 = s_2 t_1 \swarrow & & \searrow f_1 t_2 + f_2 t_1 \\ A^\bullet & & B^\bullet. \end{array}$$

There is a natural additive functor $Q_{\mathcal{A}} : K(\mathcal{A}) \rightarrow D(\mathcal{A})$, which is the identity on objects, and sends a morphism $f : A^\bullet \rightarrow B^\bullet$ to the equivalence class of the roof

$$\begin{array}{ccc} & A^\bullet & \\ \text{id} \swarrow & & \searrow f \\ A^\bullet & & B^\bullet. \end{array}$$

In fact, $D(\mathcal{A})$ has a triangulated structure, in a way compatible with $Q_{\mathcal{A}}$. The *shift functor*, which is again denoted by $[1]$, is defined on a morphism

$$A^\bullet \xleftarrow{s} C^\bullet \xrightarrow{f} B^\bullet$$

by

$$A^\bullet[1] \xleftarrow{s[1]} C^\bullet[1] \xrightarrow{f[1]} B^\bullet[1].$$

A *distinguished triangle* is defined to be a sequence in $D(\mathcal{A})$ isomorphic to the image of a distinguished triangle in $K(\mathcal{A})$ under Q , i.e. to a sequence of the form

$$A^\bullet \xrightarrow{Q(f)} B^\bullet \xrightarrow{Q(\tau)} C(f) \xrightarrow{Q(\pi)} A^\bullet[1].$$

For a proof that this specifies a triangulated structure on $D(\mathcal{A})$, see [*].

We will show next that every short exact sequence

$$0 \longrightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \longrightarrow 0 \quad (3.2)$$

in $\text{Com}(\mathcal{A})$ determines a distinguished triangle in $D(\mathcal{A})$. Note we have a factorisation

$$\begin{array}{ccc} & C(f) & \\ \tau \nearrow & \downarrow \gamma & \\ B^\bullet & \xrightarrow{g} & C^\bullet, \end{array} \quad (3.3)$$

where $\gamma^i : C(f)^i \rightarrow C^i$ is given by $(a, b) \mapsto g^i(b)$.

Proposition 3.4. *The morphism γ in (3.3) is a quasi-isomorphism. Hence, γ^{-1} exists in $D(\mathcal{A})$. Setting $\delta := \pi \circ \gamma^{-1}$, we have a commutative diagram in $D(\mathcal{A})$:*

$$\begin{array}{ccccccc} A^\bullet & \xrightarrow{f} & B^\bullet & \xrightarrow{\tau} & C(f) & \xrightarrow{\pi} & A^\bullet[1] \\ \parallel & & \parallel & & \downarrow \gamma & & \parallel \\ A^\bullet & \xrightarrow{f} & B^\bullet & \xrightarrow{g} & C^\bullet & \xrightarrow{\delta} & A^\bullet[1], \end{array}$$

which shows that the bottom row is a distinguished triangle.

Proof. We will show that the mapping cone $C(\gamma)$ is exact using a diagram chase. The differential $d_{C(\gamma)}^i : A^{i+2} \oplus B^{i+1} \oplus C^i \rightarrow A^{i+3} \oplus B^{i+2} \oplus C^{i+1}$ of $C(\gamma)$ is given by the matrix

$$d_{C(\gamma)}^i = \begin{bmatrix} d_A^{i+2} & 0 & 0 \\ -f^{i+2} & -d_B^{i+1} & 0 \\ 0 & g^{i+1} & d_C^i \end{bmatrix}.$$

If $(a, b, c) \in A^{i+2} \oplus B^{i+1} \oplus C^i$ is in the kernel of $d_{C(\gamma)}^i$, then this implies the relations

$$d_A^{i+2}(a) = 0, \quad f^{i+2}(a) = d_B^{i+1}(-b), \quad g^{i+1}(-b) = d_C^i(c),$$

which we can visualise on the short exact sequence (3.2) as follows:

$$\begin{array}{ccc} 0 & & \\ \uparrow & & \\ a & \longmapsto & f^{i+2}(a) = d_B^{i+1}(-b) \\ & & \uparrow \\ & & -b \longmapsto g^{i+1}(-b) = d_C^i(c) \\ & & \uparrow \\ & & c \end{array}$$

Since g is an epimorphism, there exists $b' \in B^i$ such that $g^i(b') = c$. Then since $-b$ and $d_B^i(b')$ both map to $d_C^i(c)$ under g^{i+1} , exactness of (3.2) at B^* implies that there exists $a' \in A^{i+1}$ such that $f^{i+1}(a') = -b - d_B^i(b')$. Now

$$f^{i+2}(d_A^{i+1}(a')) = d_B^{i+1}(f^{i+1}(a')) = d_B^{i+1}(-b) = f^{i+2}(a),$$

and since f is a monomorphism, we have $d_A^{i+1}(a') = a$. Finally, compute that $d_{C(\gamma)}^{i-1}(a', b', 0) = (a, b, c)$, and apply Lemma 2.2 to conclude the proof. $\square$

Remark 3.5. We now have two long exact cohomology sequences associated to the short exact sequence (3.2)! The terms are all the same of course, but is $H^i(\delta)$ equal to the usual connecting morphism? It turns out that it is, up to a sign. However, we will omit the verification of the point, since it requires a construction (the mapping cylinder) which we have avoided in our presentation. For details, see [*].

The connecting morphism δ is functorial in the short exact sequence:

Proposition 3.6. *A commutative diagram in $\text{Com}(\mathcal{A})$ of the form*

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_1^\bullet & \xrightarrow{f_1} & B_1^\bullet & \xrightarrow{g_1} & C_1^\bullet \longrightarrow 0 \\ & & \downarrow a & & \downarrow b & & \downarrow c \\ 0 & \longrightarrow & A_2^\bullet & \xrightarrow{f_2} & B_2^\bullet & \xrightarrow{g_2} & C_2^\bullet \longrightarrow 0 \end{array}$$

gives rise to a morphism of triangles in $\text{D}(\mathcal{A})$:

$$\begin{array}{ccccccc} A_1^\bullet & \xrightarrow{f_1} & B_1^\bullet & \xrightarrow{g_1} & C_1^\bullet & \xrightarrow{\delta_1} & A_1^\bullet \\ \downarrow a & & \downarrow b & & \downarrow c & & \downarrow a[1] \\ A_2^\bullet & \xrightarrow{f_2} & B_2^\bullet & \xrightarrow{g_2} & C_2^\bullet & \xrightarrow{\delta_2} & A_2^\bullet, \end{array}$$

where δ_1, δ_2 are the connecting morphisms of Proposition 3.4.

Proof. We already know that the first two squares commute, while the commutativity of the third square is easy to check from the definitions of δ_1 and δ_2 . $\square$

Finally, imposing boundedness conditions...

Definition 3.7. Definition of $\text{Com}^*(\mathcal{A}), K^*(\mathcal{A}), D^*(\mathcal{A})$ for $* = +, -, b$.

4. INJECTIVE OBJECTS AND RESOLUTIONS

In this section, we study injective objects in an abelian category $\mathcal{A}$. This notion will be crucial in our treatment of derived functors. More precisely, we will assume that $\mathcal{A}$ has sufficiently many injective objects in order to prove right derived functors exist.

Recall that for any object $B \in \mathcal{A}$, the contravariant functor $\text{Hom}(-, B)$ is left exact. This means that for any exact sequence

$$A_1 \longrightarrow A_2 \longrightarrow A_3 \longrightarrow 0,$$

of objects in $\mathcal{A}$, the sequence of abelian groups

$$0 \longrightarrow \text{Hom}(A_3, B) \longrightarrow \text{Hom}(A_2, B) \longrightarrow \text{Hom}(A_1, B)$$

remains exact.

Definition 4.1. An object $I \in \mathcal{A}$ is *injective* if the functor $\text{Hom}(-, I)$ is right exact (and hence exact). We let $\mathcal{I}$ denote the full subcategory of $\mathcal{A}$ consisting of injective objects. Note that $\mathcal{I}$ is an additive category, so the homotopy categories $K^*(\mathcal{I})$ are defined.

This definition has a diagrammatic reformulation. Let $I \in \mathcal{A}$ be an injective object. Suppose we are given an exact sequence

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C$$

and a morphism $\gamma : B \rightarrow I$ such that $\gamma \circ \alpha = 0$. Then there exists a morphism $\delta : C \rightarrow I$ making the diagram

$$\begin{array}{ccccc} A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C \\ & & \downarrow \gamma & \swarrow \delta & \\ & & I & & \end{array} \quad (4.1)$$

commutative. In particular, this holds if $A = 0$, so that β is a monomorphism.

Definition 4.2. We say that $\mathcal{A}$ has *enough injectives* if for every object $A \in \mathcal{A}$, there is a monomorphism $A \hookrightarrow I$, where $I \in \mathcal{A}$ is injective.

If $\mathcal{A}$ has enough injectives, then it is easy to see that every object $A \in \mathcal{A}$ admits an *injective resolution*, i.e. there is an exact complex of the form

$$0 \longrightarrow A \longrightarrow I^0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow \dots$$

where the I^p are injective. More generally, if $A^\bullet \in K^+(\mathcal{A})$ is a bounded-below complex, there exists a quasi-isomorphism $A^\bullet \rightarrow I^\bullet$, where $I^\bullet \in K^+(\mathcal{I})$ is a complex consisting of injective objects. To prove this, we will use Cartan-Eilenberg resolutions, which we introduce now.

Definition 4.3. Suppose we are given an upper half-plane double complex $(I^{\cdot,\cdot}, d_{\uparrow}, d_{\rightarrow})$ (so that $I^{p,q} = 0$ for $q < 0$), together with a map of complexes $\varepsilon : A^{\cdot} \rightarrow I^{\cdot,0}$. Then for each $p \in \mathbb{Z}$, we have the complexes

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & & \vdots \\
 & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 & I^{p,2} & & B_{\rightarrow}^p(I^{\cdot,2}) & & Z_{\rightarrow}^p(I^{\cdot,2}) & & H_{\rightarrow}^p(I^{\cdot,2}) \\
 & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 & I^{p,1} & & B_{\rightarrow}^p(I^{\cdot,1}) & & Z_{\rightarrow}^p(I^{\cdot,1}) & & H_{\rightarrow}^p(I^{\cdot,1}) \\
 & \uparrow & & \uparrow & & \uparrow & & \uparrow \\
 & I^{p,0} & & B_{\rightarrow}^p(I^{\cdot,0}) & & Z_{\rightarrow}^p(I^{\cdot,0}) & & H_{\rightarrow}^p(I^{\cdot,0}) \\
 & \varepsilon^p \uparrow & & B_{\rightarrow}^p(\varepsilon) \uparrow & & Z_{\rightarrow}^p(\varepsilon) \uparrow & & H_{\rightarrow}^p(\varepsilon) \uparrow \\
 & A^p & & B^p(A^{\cdot}) & & Z^p(A^{\cdot}) & & H^p(A^{\cdot})
 \end{array} \tag{4.2}$$

(Here $B_{\rightarrow}^p$, $Z_{\rightarrow}^p$, and $H_{\rightarrow}^p$ denote respectively coboundaries, cocycles, and cohomology taken with respect to the horizontal differential $d_{\rightarrow}$.) We say that $(I^{\cdot,\cdot}, d_{\uparrow}, d_{\rightarrow})$ is a *Cartan-Eilenberg resolution* of $A^{\cdot}$ if the following properties hold:

- i) There exists an integer p_0 such that for all $p < p_0$, the column $I^{p,\cdot}$ is the zero complex.
- ii) For all $p \in \mathbb{Z}$, the morphisms ε^p , $B_{\rightarrow}^p(\varepsilon)$, $Z_{\rightarrow}^p(\varepsilon)$ and $H_{\rightarrow}^p(\varepsilon)$ in (4.2) are injective resolutions. That is, the four complexes in (4.2) are acyclic, and the objects $I^{p,q}$, $B_{\rightarrow}^p(I^{\cdot,q})$, $Z_{\rightarrow}^p(I^{\cdot,q})$, and $H_{\rightarrow}^p(I^{\cdot,q})$ are all assumed to be injective.

To prove the existence of Cartan-Eilenberg resolutions, we need the following result, which is known as the horseshoe lemma.

Lemma 4.4. *Let $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ be a short exact sequence in $\mathcal{A}$. Given any two injective resolutions $A_1 \rightarrow I_1^{\cdot}$, $A_3 \rightarrow I_3^{\cdot}$, there exists a third injective resolution $A_2 \rightarrow I_2^{\cdot}$ fitting into a commutative diagram with exact rows*

$$\begin{array}{ccccccc}
 0 & \longrightarrow & I_1^{\cdot} & \longrightarrow & I_2^{\cdot} & \longrightarrow & I_3^{\cdot} \longrightarrow 0 \\
 & & \uparrow & & \uparrow & & \uparrow \\
 0 & \longrightarrow & A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 \longrightarrow 0.
 \end{array}$$

Theorem 4.5. *Suppose $\mathcal{A}$ has enough injectives. Then every bounded below complex $A^{\cdot} \in K^+(\mathcal{A})$ has a Cartan-Eilenberg resolution.*

Proof. We will construct the double complex $I^{\cdot,\cdot}$ recursively. For brevity, we will write $B^p := B^p(A^{\cdot})$, $Z^p := Z^p(A^{\cdot})$ and $H^p := H^p(A^{\cdot})$.

Let p_0 be the smallest integer such that $A^{p_0} \neq 0$. Then $Z^{p_0} = H^{p_0}$. Choose injective resolutions

$$Z^{p_0} \rightarrow I_Z^{p_0,\cdot}, \quad B^{p_0+1} \rightarrow I_B^{p_0+1,\cdot},$$

and apply Lemma 4.4 to the short exact sequence $0 \rightarrow Z^{p_0} \rightarrow A^{p_0} \rightarrow B^{p_0+1} \rightarrow 0$ to get a resolution $\varepsilon^{p_0} : A^{p_0} \rightarrow I^{p_0,\cdot}$ that fits into a short exact sequence

$$0 \longrightarrow I_Z^{p_0,\cdot} \longrightarrow I^{p_0,\cdot} \longrightarrow I_B^{p_0+1,\cdot} \longrightarrow 0.$$

Now let $p > p_0$, and assume for all $p' < p$ that we have chosen injective resolutions $Z^{p'} \rightarrow I_Z^{p',\bullet}$, $H^{p'} \rightarrow I_H^{p',\bullet}$, $B^{p'+1} \rightarrow I_B^{p'+1,\bullet}$ and $\varepsilon^{p'} : A^{p'} \rightarrow I^{p',\bullet}$ fitting into short exact sequences

$$0 \longrightarrow I_B^{p',\bullet} \longrightarrow I_Z^{p',\bullet} \longrightarrow I_H^{p',\bullet} \longrightarrow 0. \quad (4.3)$$

and

$$0 \longrightarrow I_Z^{p',\bullet} \longrightarrow I^{p',\bullet} \longrightarrow I_B^{p'+1,\bullet} \longrightarrow 0. \quad (4.4)$$

In particular, we have a resolution $B^p \rightarrow I_B^{p,\bullet}$. Choose any resolution $H^p \rightarrow I_H^{p,\bullet}$, and apply Lemma 4.4 to obtain an injective resolution $Z^p \rightarrow I_Z^{p,\bullet}$ fitting into the short exact sequence (4.3) with $p' = p$. Now pick a resolution $B^{p+1} \rightarrow I_B^{p+1,\bullet}$, and apply Lemma 4.4 again to get a resolution $\varepsilon^p : A^p \rightarrow I^{p,\bullet}$ fitting into the short exact sequence (4.4) with $p' = p$. This closes the recursion.

We now make $I^{\bullet,\bullet}$ into a bicomplex, and verify that it has the stated properties. We define $d_{\uparrow}$ in the p th column to be $(-1)^p$ times the differential that the complex $I^{p,\bullet}$ already possesses. The horizontal differentials $d_{\rightarrow}$ are given, out of the p th column, by the composition

$$I^{p,\bullet} \longrightarrow I_B^{p+1,\bullet} \longrightarrow I_Z^{p+1,\bullet} \longrightarrow I^{p+1,\bullet}, \quad (4.5)$$

using the maps from (4.3) and (4.4). It is then immediate that $d_{\rightarrow}^2 = 0$, and also that $d_{\rightarrow}d_{\uparrow} + d_{\uparrow}d_{\rightarrow} = 0$, so that $(I^{\bullet,\bullet}, d_{\uparrow}, d_{\rightarrow})$ is a bicomplex. Moreover, with this definition of $d_{\rightarrow}$, the map $\varepsilon : A^{\bullet} \rightarrow I^{\bullet,0}$ is a morphism of complexes; this follows from the commutativity of the diagram

$$\begin{array}{ccccccc} I^{p,0} & \longrightarrow & I_B^{p+1,0} & \longrightarrow & I_Z^{p+1,0} & \longrightarrow & I^{p+1,0} \\ \varepsilon^p \uparrow & & \uparrow & & \uparrow & & \varepsilon^{p+1} \uparrow \\ A^p & \longrightarrow & B^{p+1} & \longrightarrow & Z^{p+1} & \longrightarrow & A^{p+1}, \end{array}$$

which was a consequence of Lemma 4.4. Finally, note that the short exact sequences (4.3) and (4.4) together with the definition (4.5) imply that $B_{\rightarrow}^p(I^{\bullet,\bullet}) = I_B^{p,\bullet}$, $Z_{\rightarrow}^p(I^{\bullet,\bullet}) = I_Z^{p,\bullet}$, and $H_{\rightarrow}^p(I^{\bullet,\bullet}) = I_H^{p,\bullet}$. $\square$

The next proposition explains why Cartan-Eilenberg resolutions are *resolutions*.

Proposition 4.6. *Suppose $I^{\bullet,\bullet}$ is a Cartan-Eilenberg resolution of a bounded-below complex $A^{\bullet} \in K^+(\mathcal{A})$. Then the totalisation $\text{Tot } I^{\bullet,\bullet}$ is a complex consisting of injective objects, and the augmentation $\varepsilon : A^{\bullet} \rightarrow I^{\bullet,0}$ induces a quasi-isomorphism $A^{\bullet} \rightarrow \text{Tot } I^{\bullet,\bullet}$.*

Proof. For simplicity, we assume that $A^p = 0$ for $p < 0$, so that $I^{\bullet,\bullet}$ is a first quadrant double complex. The general case can then be easily obtained by shifting $A^{\bullet}$ to the right.

It is clear that $\text{Tot } I^{\bullet,\bullet}$ consists of injective objects, since $\text{Tot}^n I^{\bullet,\bullet} = \bigoplus_{p+q=n} I^{p,q}$, and finite direct sums of injective objects are injective. To see that ε induces a morphism of complexes $A^{\bullet} \rightarrow \text{Tot } I^{\bullet,\bullet}$, note that ε identifies $A^{\bullet}$ with the subcomplex $\text{Ker } d_{\uparrow}^{\bullet,0}$ of $I^{\bullet,0}$, and that the natural inclusion $\text{Ker } d_{\uparrow}^{\bullet,0} \rightarrow \text{Tot } I^{\bullet,\bullet}$ is a morphism of complexes (recall that the differential on $\text{Tot } I^{\bullet,\bullet}$ is given by $d_{\uparrow} + d_{\rightarrow}$).

To show that the map $A^\bullet \rightarrow \text{Tot } I^{\bullet,\bullet}$ is a quasi-isomorphism, we will make use of the rightward-oriented spectral sequence associated to the first quadrant double complex $I^{\bullet,\bullet}$. The differential on the 0th page is $d_{\rightarrow}$, so the 1st page looks like

$$\begin{array}{ccccc}
 & \vdots & & \vdots & & \vdots \\
 & \uparrow & & \uparrow & & \uparrow \\
 & H_{\rightarrow}^{p-1}(I^{\bullet,2}) & & H_{\rightarrow}^p(I^{\bullet,2}) & & H_{\rightarrow}^{p+1}(I^{\bullet,2}) \\
 & \uparrow & & \uparrow & & \uparrow \\
 & H_{\rightarrow}^{p-1}(I^{\bullet,1}) & & H_{\rightarrow}^p(I^{\bullet,1}) & & H_{\rightarrow}^{p+1}(I^{\bullet,1}) \\
 & \uparrow & & \uparrow & & \uparrow \\
 & H_{\rightarrow}^{p-1}(I^{\bullet,0}) & & H_{\rightarrow}^p(I^{\bullet,0}) & & H_{\rightarrow}^{p+1}(I^{\bullet,0}),
 \end{array}$$

where the differentials are induced by $d_{\uparrow}$. Now since $I^{\bullet,\bullet}$ is a Cartan-Eilenberg resolution, $H_{\rightarrow}^p(I^{\bullet,\bullet})$ is an injective resolution of $H^p(A^\bullet)$, so the 2nd page looks like

$$\begin{array}{ccccc}
 & \vdots & & \vdots & & \vdots \\
 & \swarrow & & \swarrow & & \swarrow \\
 0 & & 0 & & 0 \\
 & \swarrow & & \swarrow & & \swarrow \\
 0 & & 0 & & 0 \\
 & \swarrow & & \swarrow & & \swarrow \\
 H^{p-1}(A^\bullet) & & H^p(A^\bullet) & & H^{p+1}(A^\bullet),
 \end{array}$$

where the differentials are all zero. The spectral sequence has thus degenerated here. Moreover, since there is at most one nonzero term in each antidiagonal, the filtration on the cohomology of $\text{Tot } I^{\bullet,\bullet}$ is trivial, and we obtain isomorphisms $H^n(A^\bullet) \cong H^n(\text{Tot } I^{\bullet,\bullet})$ for all n . The fact that these isomorphisms are induced by the map $A^\bullet \rightarrow \text{Tot } I^{\bullet,\bullet}$ follows from the construction of the spectral sequence. $\square$

In the remainder of this section, we discuss maps to complexes of injectives $I^\bullet$.

Lemma 4.7. *Suppose $q : B^\bullet \rightarrow A^\bullet$ is a quasi-isomorphism in $K^+(\mathcal{A})$. For any complex of injective objects $I^\bullet \in K^+(\mathcal{A})$, the induced map*

$$\text{Hom}_{K^+(\mathcal{A})}(A^\bullet, I^\bullet) \xrightarrow{- \circ q} \text{Hom}_{K^+(\mathcal{A})}(B^\bullet, I^\bullet)$$

is an isomorphism.

Proof. Let $q : B^\bullet \rightarrow A^\bullet$ denote the given quasi-isomorphism, and consider the distinguished triangle in $K^+(\mathcal{A})$:

$$B^\bullet \xrightarrow{q} A^\bullet \longrightarrow C(q) \longrightarrow B^\bullet[1].$$

By Lemma 2.2, the mapping cone $C(q)$ is acyclic. Moreover, since $\text{Hom}_{K^+(\mathcal{A})}(-, I^\bullet)$ is a homological functor (see Example 2.9), this triangle gives rise to a long exact

sequence as in (2.2), a piece of which is

$$\begin{array}{c} \text{Hom}(C(q)[-1], I^\bullet) \\ \curvearrowright \\ \text{Hom}(C(q), I^\bullet) \longrightarrow \text{Hom}(A^\bullet, I^\bullet) \xrightarrow{-\circ q} \text{Hom}(B^\bullet, I^\bullet) \end{array}$$

Hence, to show that the map $-\circ q$ is an isomorphism, it will be enough to show that $\text{Hom}(C^\bullet, I^\bullet) = 0$ for any acyclic complex $C^\bullet \in \mathbf{K}^+(\mathcal{A})$. Let $g : C^\bullet \rightarrow I^\bullet$ be a map of complexes. We claim g is homotopy equivalent to zero, i.e. there exist morphisms $h^n : C^n \rightarrow I^{n-1}$ such that $hd + dh = g$. We will construct h using induction on n . For convenience, we may assume $C^n = 0$ for all $n < 0$. Consider the diagram

$$\begin{array}{ccccccc} \cdots & \longrightarrow & 0 & \longrightarrow & C^0 & \longrightarrow & C^1 \longrightarrow \cdots \\ & & \downarrow g^{-1} & & \downarrow g^0 & & \downarrow g^1 \\ \cdots & \longrightarrow & I^{-1} & \longrightarrow & I^0 & \longrightarrow & I^1 \longrightarrow \cdots \end{array}$$

We set $h^n = 0$ for all $n < 0$. Since I^0 is injective and $C^\bullet$ is exact in degree 0, we can use the lifting property (4.1) to get a map $h^1 : C^1 \rightarrow I^0$ such that $g^0 = h^1 d^0$. Now let $n \geq 0$, and assume we have constructed maps $h^i : C^i \rightarrow I^{i-1}$ such that

$$g^{i-1} = h^i d^{i-1} + d^{i-2} h^{i-1} \quad (4.6)$$

for all $i \leq n$. Consider the diagram

$$\begin{array}{ccccc} C^{n-1} & \xrightarrow{d^{n-1}} & C^n & \xrightarrow{d^n} & C^{n+1} \\ & & \downarrow g^n - d^{n-1} h^n & & \\ & & I^n & & \end{array}$$

A quick calculation verifies that $(g^n - d^{n-1} h^n) \circ d^{n-1} = 0$, so the lifting property (4.1) implies that there exists a morphism $h^{n+1} : C^{n+1} \rightarrow I^n$ satisfying (4.6) with $i = n + 1$. $\square$

The following theorem will be our main tool for computing in the derived category:

Theorem 4.8. *Let $A^\bullet \in \mathbf{K}^+(\mathcal{A})$ be a bounded-below complex, and $I^\bullet \in \mathbf{K}^+(\mathcal{I})$ a bounded-below complex of injectives. Then the natural map*

$$Q_{\mathcal{A}} : \text{Hom}_{\mathbf{K}^+(\mathcal{A})}(A^\bullet, I^\bullet) \longrightarrow \text{Hom}_{\mathbf{D}^+(\mathcal{A})}(A^\bullet, I^\bullet)$$

is an isomorphism.

Proof. First, we will prove that $Q_{\mathcal{A}}$ is injective. Let $f, g \in \text{Hom}_{\mathbf{K}^+(\mathcal{A})}(A^\bullet, I^\bullet)$ such that $Q_{\mathcal{A}}(f) = Q_{\mathcal{A}}(g)$. Then the roofs

$$A^\bullet \xleftarrow{\text{id}} A^\bullet \xrightarrow{f} I^\bullet, \quad A^\bullet \xleftarrow{\text{id}} A^\bullet \xrightarrow{g} I^\bullet,$$

are equivalent, so there exists a quasi-isomorphism $q : B^\bullet \rightarrow A^\bullet$ in $\mathbf{K}^+(\mathcal{A})$ such that $f \circ q = g \circ q$. By Lemma 4.7, we have $f = g$.

To prove that $Q_{\mathcal{A}}$ is surjective, we have to show that for every roof

$$\begin{array}{ccc} & B^\bullet & \\ q \swarrow & & \searrow f \\ A^\bullet & & I^\bullet, \end{array}$$

there exists a morphism of complexes $g : A^\bullet \rightarrow I^\bullet$ such that $g \circ q = f$ in $K^+(\mathcal{A})$. Again, this follows from Lemma 4.7. $\square$

Let ι denote the restriction of the functor $Q_{\mathcal{A}}$ to the subcategory $K^+(\mathcal{I})$.

Theorem 4.9. *If $\mathcal{A}$ has enough injectives, the natural functor*

$$\iota : K^+(\mathcal{I}) \longrightarrow D^+(\mathcal{A})$$

is an equivalence of triangulated categories.

Proof. Since $\mathcal{A}$ has enough injectives, Theorem 4.5 and Proposition 4.6 imply that every object $A^\bullet \in D^+(\mathcal{A})$ is isomorphic to a complex of injective objects. Hence, ι is essentially surjective. The fact that ι is fully faithful follows from Theorem 4.8. $\square$

Corollary 4.10. *If $\mathcal{A}$ has enough injectives, the derived category $D^+(\mathcal{A})$ exists as a category in our universe.*

5. EXISTENCE OF RF

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, and let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a left exact additive functor. The right derived functor $RF : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ of F is a triangulated functor satisfying a certain universal property. The main business of this section is to spell out this universal property, and to prove that RF exists if $\mathcal{A}$ has enough injective objects. (The assumption that F is left exact is not strictly necessary here, in that one can define RF without it. However, there is no reason for RF to be meaningfully related to F in this case.)

Any additive functor $F : \mathcal{A} \rightarrow \mathcal{B}$ induces a functor $K(F) : K^*(\mathcal{A}) \rightarrow K^*(\mathcal{B})$, which maps the complex

$$\dots \xrightarrow{d^{i-2}} A^{i-1} \xrightarrow{d^{i-1}} A^i \xrightarrow{d^i} A^{i+1} \xrightarrow{d^{i+1}} \dots$$

to the complex

$$\dots \xrightarrow{Fd^{i-2}} FA^{i-1} \xrightarrow{Fd^{i-1}} FA^i \xrightarrow{Fd^i} FA^{i+1} \xrightarrow{Fd^{i+1}} \dots$$

This functor is *triangulated*, in the sense of the following definition:

Definition 5.1. Let $\mathcal{T}_1, \mathcal{T}_2$ be triangulated categories. An additive functor $\mathcal{T}_1 \rightarrow \mathcal{T}_2$ is *triangulated* if it commutes with the shift functors (up to a natural isomorphism which is part of the data) and maps distinguished triangles to distinguished triangles.

Consider the problem of completing the following diagram to a commutative square:

$$\begin{array}{ccc} K(\mathcal{A}) & \xrightarrow{K(F)} & K(\mathcal{B}) \\ \downarrow Q_{\mathcal{A}} & & \downarrow Q_{\mathcal{B}} \\ D(\mathcal{A}) & \dashrightarrow & D(\mathcal{B}), \end{array} \tag{5.1}$$

where the dashed arrow represents a triangulated functor. If F is an exact functor, then using the mapping cone construction one easily sees that $K(F)$ preserves quasi-isomorphisms, and hence descends to a triangulated functor $D(\mathcal{A}) \rightarrow D(\mathcal{B})$. This functor is the *derived functor* of F .

If F is not exact, then its derived functor is something much more interesting than this. Completing (5.1) to a commutative square is now no longer possible. In what follows, we will consider the case when F is exact on the left. It is then appropriate to restrict attention to the bounded-below categories $K^+(\mathcal{A})$, $D^+(\mathcal{A})$, etc.

Definition 5.2. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor which is left exact. The *right derived functor* of F is a triangulated functor

$$RF : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{B})$$

together with a natural transformation

$$\zeta_F : Q_{\mathcal{B}} \circ K(F) \longrightarrow RF \circ Q_{\mathcal{A}}.$$

Together, they satisfy the following universal property: for any triangulated functor $G : D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$ and any natural transformation $\zeta : Q_{\mathcal{B}} \circ K(F) \rightarrow G \circ Q_{\mathcal{A}}$, there exists a unique natural transformation $\theta : RF \rightarrow G$ making the diagram

$$\begin{array}{ccc} Q_{\mathcal{B}} \circ K(F) & \xrightarrow{\zeta_F} & RF \circ Q_{\mathcal{A}} \\ & \searrow \zeta & \downarrow \theta \circ \text{id}_{Q_{\mathcal{A}}} \\ & & G \circ Q_{\mathcal{A}} \end{array} \quad (5.2)$$

commute.

We now construct RF , assuming that $\mathcal{A}$ has enough injective objects. By Theorem 4.9, we have an equivalence of triangulated categories $\iota : K^+(\mathcal{I}) \rightarrow D^+(\mathcal{A})$, which is evidently triangulated. Choose a quasi-inverse ι^{-1} to ι (which is then automatically triangulated). We may assume that ι^{-1} is the identity on objects of $K^+(\mathcal{I})$. Consider the diagram

$$\begin{array}{ccccc} K^+(\mathcal{I}) & \hookrightarrow & K^+(\mathcal{A}) & \xrightarrow{K(F)} & K^+(\mathcal{B}) \\ & \searrow \iota & \downarrow Q_{\mathcal{A}} & & \downarrow Q_{\mathcal{B}} \\ & \swarrow \iota^{-1} & D^+(\mathcal{A}) & & D^+(\mathcal{B}), \end{array}$$

where $K^+(\mathcal{I}) \hookrightarrow K^+(\mathcal{A})$ is the canonical embedding. Our candidate for the right derived functor of F is

$$RF := Q_{\mathcal{B}} \circ K(F) \circ \iota^{-1}.$$

Note that RF is triangulated, since it is a composition of triangulated functors.

To construct ζ_F , consider the functor $\iota^{-1} \circ Q_{\mathcal{A}}$, which sends an object $A^\bullet \in K^+(\mathcal{A})$ to the chosen injective resolution $I^\bullet := \iota^{-1}(A^\bullet) \in K^+(\mathcal{A})$. We claim there is a natural transformation of functors on $K^+(\mathcal{A})$

$$\eta : \text{id}_{K^+(\mathcal{A})} \longrightarrow \iota^{-1} \circ Q_{\mathcal{A}}.$$

Since ι^{-1} is a quasi-inverse to ι , we have a natural isomorphism $\alpha : \text{id}_{D^+(\mathcal{A})} \rightarrow \iota \circ \iota^{-1}$. By Theorem 4.8, that the natural map

$$\text{Hom}_{K^+(\mathcal{A})}(A^\bullet, I^\bullet) \longrightarrow \text{Hom}_{D^+(\mathcal{A})}(A^\bullet, I^\bullet) \quad (5.3)$$

is an isomorphism, so we may define η_A to be the preimage of α_A under (5.3); note that η_A is a quasi-isomorphism. The naturality of η is immediate. Postcomposing η with the identity transformation of $Q_B \circ K(F)$ defines ζ_F .

Remark 5.3. For all $A^\bullet, B^\bullet \in K^+(\mathcal{A})$, we have natural isomorphisms

$$\begin{aligned} \mathrm{Hom}_{D^+(\mathcal{A})}(Q_{\mathcal{A}}(A^\bullet), B^\bullet) &\cong \mathrm{Hom}_{D^+(\mathcal{A})}(Q_{\mathcal{A}}(A^\bullet), Q_{\mathcal{A}}(\iota^{-1}(B^\bullet))) \\ &\cong \mathrm{Hom}_{K^+(\mathcal{A})}(A^\bullet, \iota^{-1}(B^\bullet)), \end{aligned}$$

where the first isomorphism is given by postcomposition with α_B , and the second is given by (5.3). This shows that $Q_{\mathcal{A}}$ is left adjoint to ι^{-1} , and η is the unit of this adjunction. Moreover, the counit ε is just α^{-1} .

Theorem 5.4. *If $\mathcal{A}$ has enough injectives, the pair RF, ζ_F defined above is the right derived functor of F .*

Proof. Given a triangulated functor G and a natural transformation ζ as in Definition 5.2, define θ to be the following composition of natural transformations:

$$RF = Q_B \circ K(F) \circ \iota^{-1} \xrightarrow{\zeta \circ \mathrm{id}_{\iota^{-1}}} G \circ Q_{\mathcal{A}} \circ \iota^{-1} \xrightarrow{\mathrm{id}_G \circ \varepsilon} G, \quad (5.4)$$

where ε is the counit from the adjunction of Remark 5.3. To show that θ makes (5.2) commute, we compute that

$$Q_B \circ K(F) \xrightarrow{\zeta_F} RF \circ Q_{\mathcal{A}} \xrightarrow{\theta \circ \mathrm{id}_{Q_{\mathcal{A}}}} G \circ Q_{\mathcal{A}}$$

is given on an object $A^\bullet \in D^+(\mathcal{A})$ by the composition of arrows

$$\begin{array}{ccc} Q_B(K(F)(A^\bullet)) & \xrightarrow{Q_B(K(F)(\eta_A))} & Q_B(K(F)(I^\bullet)) \\ & & \downarrow \zeta_I \\ G(A^\bullet) & \xleftarrow{G(\varepsilon_I)} & G(I^\bullet), \end{array}$$

where $I^\bullet := \iota^{-1}(A^\bullet)$. But $\varepsilon_I = Q_{\mathcal{A}}(\eta_A)^{-1}$ and ζ is a natural transformation, so this composition is just equal to ζ_A .

For the uniqueness part of the statement, suppose $\tilde{\theta}$ is a natural transformation making (5.2) commute. By naturality, $\tilde{\theta}_A$ factorises as

$$\tilde{\theta}_A = G(\varepsilon_I) \circ \tilde{\theta}_I \circ RF((\varepsilon_A)^{-1}).$$

Since ι^{-1} is the identity on objects of $K^+(\mathcal{I})$, $(\zeta_F)_I$ is the identity transformation of $Q_B \circ K(F)(I^\bullet) = RF(I^\bullet)$, and the commutativity of (5.2) then implies that $\zeta_I = \tilde{\theta}_I$ as morphisms $RF(I^\bullet) \rightarrow G(I^\bullet)$. This shows $\tilde{\theta}_A = \theta_A$. $\square$

6. PROPERTIES OF RF

We continue with the setup from the previous section: $\mathcal{A}$ is an abelian category with enough injectives, $\mathcal{B}$ is an abelian category, and $F : \mathcal{A} \rightarrow \mathcal{B}$ is an additive functor which is left exact. In this last section, we establish some important properties of the right derived functor RF . We are especially interested in the *higher derived functors* of F . These functors encompass much of classical homological algebra, and the technology of the derived category simplifies their theory substantially.

The higher derived functors are defined on the abelian category $\mathcal{A}$, so we have to embed $\mathcal{A}$ into $D^+(\mathcal{A})$ somehow. There is a standard way to do this:

Lemma 6.1. *Considering an object $A \in \mathcal{A}$ as a complex concentrated in degree 0 realises $\mathcal{A}$ as a full subcategory of $D^+(\mathcal{A})$.*

From now on, we will identify an object $A \in \mathcal{A}$ with its image under the above embedding into $D^+(\mathcal{A})$ without further comment.

Definition 6.2. Let $i \geq 0$ be an integer. Setting

$$R^i F(A) := H^i(RF(A))$$

defines an additive functor $R^i F : \mathcal{A} \rightarrow \mathcal{B}$, called the i th *higher derived functor* of F .

To compute $R^i F(A)$, note that $\iota^{-1}(A)$ is an injective resolution of A :

$$0 \longrightarrow A \longrightarrow I^0 \longrightarrow I^1 \longrightarrow \cdots,$$

and that $R^i F(A)$ is the i th cohomology of the complex $F(I^\bullet) := K(F)(I^\bullet)$. Our assumption that F is left exact implies the crucial fact that $R^0 F(A) \cong F(A)$.

By Proposition 3.4, any short exact sequence in $\mathcal{A}$ determines a distinguished triangle in $D^+(\mathcal{A})$. Since the functor RF is triangulated, the image of this triangle is a distinguished triangle in $D^+(\mathcal{B})$. Consideration of the corresponding long exact cohomology sequence (2.2) then yields the following amazing result:

Theorem 6.3. *Every short exact sequence in $\mathcal{A}$:*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0,$$

gives rise to a long exact sequence in $\mathcal{B}$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(A) & \longrightarrow & F(B) & \longrightarrow & F(C) \\ & & & & \searrow^{F(\delta)} & & \nearrow \\ & & R^1 F(A) & \longrightarrow & R^1 F(B) & \longrightarrow & R^1 F(C) \\ & & & & \searrow^{R^1 F(\delta)} & & \nearrow \\ & & R^2 F(A) & \longrightarrow & R^2 F(B) & \longrightarrow & \cdots \end{array} \quad (6.1)$$

where $\delta : C \rightarrow A[1]$ is the connecting morphism of Proposition 3.4. Moreover, the long exact sequence (6.1) is functorial in the short exact sequence.

Proof. The existence of (6.1) is clear from the above remarks. The functoriality statement follows from Proposition 3.6. $\square$

Conceptually, the higher derived functors $R^i F$ measure the failure of F to be exact on the right, and Theorem 6.3 explains the precise sense in which this intuition is true.

Next, we will discuss how the higher derived functors are computed in practice. Unfortunately, in most abelian categories $\mathcal{A}$, the injective objects are not so easy to get a handle on, and while we have repeatedly seen that injective resolutions are a useful theoretical tool, one cannot usually write them down explicitly. It is necessary to allow other types of resolutions, which may be adapted to the functor F .

Definition 6.4. An object $X \in \mathcal{A}$ is said to be *F -acyclic* if $R^i F(X) = 0$ for all $i > 0$.

For example, any injective object $I \in \mathcal{A}$ is F -acyclic.

Lemma 6.5. *If $X^\bullet$ is an exact complex of F -acyclic objects, the complex $F(X^\bullet)$ remains exact.*

Proof. First note that if $X^\bullet$ is a short exact sequence of the form

$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0,$$

then the exactness of $F(X^\bullet)$ follows from the long exact sequence (6.1) and the vanishing of $R^1F(X')$. The general case follows by breaking up $X^\bullet$ into short exact sequences. $\square$

Definition 6.6. If $A \in \mathcal{A}$ is any object, then an F -acyclic resolution of A is an exact complex of the form

$$0 \longrightarrow A \longrightarrow X^0 \longrightarrow X^1 \longrightarrow X^2 \longrightarrow \dots$$

where the X^i are F -acyclic.

The next theorem says that the higher derived functors R^iF can be computed using F -acyclic resolutions.

Theorem 6.7. *For every F -acyclic resolution $A \rightarrow X^\bullet$ in $\mathcal{A}$, we have a canonical isomorphism*

$$H^i(F(X^\bullet)) \xrightarrow{\sim} R^iF(A).$$

For every commutative square of F -acyclic resolutions

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow \text{qis} & & \downarrow \text{qis} \\ X^\bullet & \xrightarrow{\tilde{f}} & Y^\bullet, \end{array}$$

the following square commutes:

$$\begin{array}{ccc} H^i(F(X^\bullet)) & \xrightarrow{H^i(F(\tilde{f}))} & H^i(F(Y^\bullet)) \\ \downarrow \sim & & \downarrow \sim \\ R^iF(A) & \xrightarrow{R^iF(f)} & R^iF(B). \end{array}$$

Proof. If we let $I^\bullet := \iota^{-1}(A)$ be the chosen injective resolution of A , then we have a roof

$$\begin{array}{ccc} & A & \\ \text{qis} \swarrow & & \searrow \eta_A \\ X^\bullet & & I^\bullet, \end{array}$$

which determines a morphism $X^\bullet \rightarrow I^\bullet$ in the derived category $D^+(\mathcal{A})$. By Theorem 4.8, this is given by an honest morphism $q : X^\bullet \rightarrow I^\bullet$ in $K^+(\mathcal{A})$.

We want to show that $F(q)$ is a quasi-isomorphism. By Lemma 2.2, the mapping cone $C(q)$ is exact. Since finite direct sums of F -acyclic objects are F -acyclic, $C(q)$ consists of F -acyclic objects, and Lemma 6.5 then implies $F(C(q))$ is exact. But $F(C(q))$ is the mapping cone of $F(q)$, so $F(q)$ is a quasi-isomorphism by the other implication of Lemma 2.2. Taking cohomology then gives the desired isomorphism $H^i(F(q)) : H^i(F(X^\bullet)) \cong R^iF(A)$.

The functoriality statement is clear from the constructions. $\square$

We now briefly discuss the Ext functor.

Definition 6.8. Recall that for any object $A \in \mathcal{A}$, the functor $\text{Hom}(A, -) : \mathcal{A} \rightarrow \mathbf{Ab}$ is left exact. We denote the i th higher derived functor of $\text{Hom}(A, -)$ by

$$\text{Ext}^i(A, -) := R^i\text{Hom}(A, -).$$

The observation we want to make here is that the Ext groups can be described in terms of homomorphism groups within the derived category:

Proposition 6.9. *There are natural isomorphisms*

$$\text{Ext}^i(A, B) \cong \text{Hom}_{\mathbf{D}(\mathcal{A})}(A, B[i])$$

for all $A, B \in \mathcal{A}$.

Proof. Let $I^\bullet := \iota^{-1}(B)$ denote the chosen injective resolution of B . By definition, $\text{Ext}^i(A, B)$ is the cohomology of the complex

$$\text{Hom}(A, I^{i-1}) \xrightarrow{d_I^{i-1} \circ -} \text{Hom}(A, I^i) \xrightarrow{d_I^i \circ -} \text{Hom}(A, I^{i+1}).$$

Now a map $f : A \rightarrow I^i$ satisfies $d_I^i \circ f = 0$ if and only if it defines a morphism of complexes $A \rightarrow I^\bullet[i]$. On the other hand, a map $g : A \rightarrow I^{i-1}$ satisfies $f = d_I^{i-1} \circ g$ if and only if it defines a homotopy between f and the zero morphism. We conclude that $\text{Ext}^i(A, B) \cong \text{Hom}_{\mathbf{K}(\mathcal{A})}(A, I^\bullet[i])$. But then by Theorem 4.8, we have

$$\text{Hom}_{\mathbf{K}(\mathcal{A})}(A, I^\bullet[i]) \cong \text{Hom}_{\mathbf{D}(\mathcal{A})}(A, I^\bullet[i]) \cong \text{Hom}_{\mathbf{D}(\mathcal{A})}(A, B[i]).$$

□

Remark 6.10. From this proposition, we see that there is a natural composition for Ext groups:

$$\text{Ext}^i(A, B) \times \text{Ext}^j(B, C) \longrightarrow \text{Ext}^{i+j}(A, C),$$

where $A, B, C \in \mathcal{A}$. Namely, an element of $\text{Ext}^i(A, B) \cong \text{Hom}_{\mathbf{D}(\mathcal{A})}(A, B[i])$ can be composed with an element of $\text{Ext}^j(B, C) \cong \text{Hom}_{\mathbf{D}(\mathcal{A})}(B[i], C[i+j])$ to produce an element of $\text{Ext}^i(A, C) \cong \text{Hom}_{\mathbf{D}(\mathcal{A})}(A, C[i+j])$. Associativity of this operation follows immediately from the categorical nature of the construction. In particular,

$$\text{Ext}^\bullet(A, A) := \bigoplus_{i \geq 0} \text{Ext}^i(A, A)$$

has the structure of a graded ring.

Finally, we will discuss compositions of derived functors. Suppose that $F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{C}$ are left exact additive functors between abelian categories. We assume that $\mathcal{A}$ and $\mathcal{B}$ have enough injectives, so that the right derived functors RF , RG , and $R(G \circ F)$ exist.

Theorem 6.11. *If F sends injectives to G -acyclic objects, then there is a canonical natural isomorphism $R(G \circ F) \cong RG \circ RF$.*

Proof. Let ζ be the natural transformation

$$Q_C \circ K(G \circ F) \xrightarrow{\zeta_{G \circ \text{id}_{\mathbf{K}(F)}}} RG \circ Q_B \circ K(F) \xrightarrow{\text{id}_{RG} \circ \zeta_F} RG \circ RF \circ Q_A.$$

By the universal property of $R(G \circ F)$, there exists a canonical natural transformation $\theta : R(G \circ F) \rightarrow RG \circ RF$, which is given as in (5.4) by

$$R(G \circ F) \xrightarrow{\zeta_{\text{id}_{\iota^{-1}}}} RG \circ RF \circ Q_A \circ \iota^{-1} \xrightarrow{\text{id}_{RG \circ RF} \circ \varepsilon} RG \circ RF.$$

This θ is simpler than it looks. Writing everything out, we see that it is a composition of two transformations: the first is

$$\zeta_G \circ \text{id}_{K(F)} \circ \text{id}_{\iota^{-1}} : R(G \circ F) \longrightarrow RG \circ Q_B \circ K(F) \circ \iota^{-1} = RG \circ RF$$

while the second is id_{RG} applied to

$$Q_B \circ K(F) \circ \iota^{-1} \xrightarrow{\zeta_F \circ \text{id}_{\iota^{-1}}} RF \circ Q_A \circ \iota^{-1} \xrightarrow{\text{id}_{RF} \circ \varepsilon} RF.$$

Now recall that ζ_F was defined to be the composite of η (the unit of the adjunction of Remark 5.3) and the identity transformation of $Q_B \circ K(F)$. So this second transformation is actually obtained by applying $\text{id}_{Q_B} \circ \text{id}_{K(F)}$ to the composite

$$\iota^{-1} \xrightarrow{\eta \circ \iota^{-1}} \iota^{-1} \circ Q_A \circ \iota^{-1} \xrightarrow{\text{id}_{\iota^{-1}} \circ \varepsilon} \iota^{-1}.$$

But this is equal to the identity on ι^{-1} , by one of the unit-counit equations. Therefore, θ is just the first transformation:

$$\theta = \zeta_G \circ \text{id}_{K(F)} \circ \text{id}_{\iota^{-1}}.$$

It remains to show that θ is a natural isomorphism. Note that if $A^\bullet \in D^+(\mathcal{A})$, we have

$$R(G \circ F)(A^\bullet) = (Q_C \circ K(G \circ F))(I^\bullet) = (Q_C \circ K(G))(X^\bullet)$$

where $I^\bullet := \iota^{-1}(A^\bullet)$ and $X^\bullet := K(F)(I^\bullet)$. What we have to show then is that the map

$$(\zeta_G)_X : (Q_C \circ K(G))(X^\bullet) \longrightarrow RG(X^\bullet)$$

is an isomorphism. But since $X^\bullet$ is a complex of G -acyclic objects, we see from the proof of Theorem 6.7 that if $J^\bullet \in K^+(\mathcal{I}_B)$ is the complex of injective objects that has been chosen to compute $X^\bullet$, and $q : X^\bullet \rightarrow J^\bullet$ is the given quasi-isomorphism, then $K(G)(q)$ remains a quasi-isomorphism. By the naturality of ζ_G , the diagram

$$\begin{array}{ccc} (Q_C \circ K(G))(X^\bullet) & \xrightarrow{(\zeta_G)_X} & RG(X^\bullet) \\ \downarrow (Q_C \circ K(G))(q) & & \downarrow RG(q) \\ (Q_C \circ K(G))(J^\bullet) & \xrightarrow{(\zeta_G)_J} & RG(J^\bullet) \end{array}$$

commutes. But $(\zeta_G)_J$ is the identity on $(Q_C \circ K(G))(J^\bullet) = RG(J^\bullet)$, and both vertical arrows are isomorphisms, so $(\zeta_G)_X$ is an isomorphism, as needed. $\square$